Shaheen Raqueeb

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.96
vidhyanjali academy	Class XII (CBSE)	2019	77.40%
kamla nehru institute of child education	Class X (CBSE)	2017	10.00%

PROJECTS

Middlebox Detection via Differential Probing | Github

- Achieved **99% detection accuracy** identifying hidden middleboxes across **3 types** using **1000 packet** dual probe trains.
- Extended a **NOMS 2024 paper** to chained middleboxes, discovering **2-3 boxes mask each other**, collapsing the detection signal from **0.50 to 0.**
- Disproved TTL based localization, showing only **3% of ICMP probes** return before the middlebox.
- Simulated real-world traffic shaping and prioritization using **tc** and **iptables** based middlebox emulation.

Dynamic Network Topology Discovery & Visualization (Ongoing) | Github

- **Designed** a CDP/LLDP parsing pipeline to extract switch neighbors, normalize device and interface information, and reconstruct network connectivity with bidirectional link deduplication.
- **Implemented** multi-switch topology visualization with interface-level link information, supporting Core, Distribution, and Access switch relationships.
- **Developed** an SSH-based collector for automated CDP/LLDP data collection and validated the pipeline using real Cisco Core and Distribution switch captures.

TransactiWar : Secure Web Application & Offensive Security Assessment | Github

- Engineered a **race condition proof** transfer engine in PHP,MySQL, guaranteeing **zero overdrafts** via **SERIALIZABLE transactions.**
- Defended against **5+ attack** classes using salted password hashing, **CSRF tokens**, and **8 tiered rate limits.**
- Hardened sessions with **HttpOnly** and **SameSite=Strict** cookies and dual timeouts, preventing **hijacking and fixation.**
- Containerized the app with **Docker Compose**, isolating web and database layers for reproducible deployment.

Agentic RAG Customer Support System | Github

- Built an agentic Q&A pipeline with **LangGraph** and **FastAPI**, routing customer queries across FAQ, order-status, refund and off-topic workflows with input/output validation.
- **Implemented** FAISS-based retrieval with BM25+RRF hybrid search and BGE reranking, achieving a **99.3% top-1 retrieval rate** at **11 ms/query** on **1,169 held-out questions**.

- **Secured** sensitive refund actions through LangGraph human-in-the-loop approval and API-key authentication, preventing **14/14 unauthorized refund attempts.**
- **Hardened** the pipeline with parallel LLM Guard scanners for prompt injection, toxicity and token limits plus output validation, backed by **45 automated tests.**

SKILLS

- **Languages:** C++, Python, SQL
- **Libraries:** NumPy, Pandas, Matplotlib, Seaborn
- **GenAI / LLM Tools :** LangGraph, LangChain, FAISS, FastAPI
- **Network Tool :** Wireshark, Mininet, TCP Dump
- **Tools and Platforms :** VS Code, Git, Linux, Docker

RELEVANT COURSES

Computer Science: Data Structures & Algorithms, Operating Systems, Advance Computer Networks, Database Management Systems, Network Security, Network Wireless System

Machine Learning: Foundations of Machine Learning, Deep Learning, Represented Learning

POSITIONS OF RESPONSIBILITY

Management Lead | HACKED, Jamia Hamdard

- Spearheaded "Hacked-a-thon", a national hackathon that drew 30+ institutes across India
- Coordinated logistics and stakeholders end to end for smooth event-day operations

Core Team Member | Literary Club, Jamia Hamdard

- Organized literary events as part of the management core team, driving campus-wide participation
- Managed logistics and participant engagement across the club's events

EXTRACURRICULAR

- **Cricket:** Winner, Inter-School Cricket Tournament, Jamia Hamdard and Winner, Milan Inter-Hostel Cricket Tournament and Bronze Medalist, Inter-department Cricket Tournament "Diesta", IIT Hyderabad.
- **Powerlifting :** Secured Silver Medal in Powerlifting in Inter-department Powerlifting Tournament "Diesta" at IIT Hyderabad
- **Cooking :** Enjoy cooking Indian and Turkish cuisine